

Lab: Case Study Analysis — Self-Model Disruptions

Objective

Analyze clinical cases involving disruptions of the self-model, applying the Active Inference framework to interpret interoceptive and body ownership phenomena.

Part 1: The Rubber Hand Illusion

Goal: Interpret a classic experimental paradigm through Active Inference.

Case: In a standard rubber hand illusion setup, participant J.K. watches a rubber hand being stroked while her own hand is hidden and stroked simultaneously. After 90 seconds, J.K. reports that the rubber hand "feels like my hand." When the experimenter unexpectedly strikes the rubber hand with a hammer, J.K. flinches and reports a brief flash of fear.

- Why does the flinch response occur even though J.K. knows the rubber hand is not hers?
- Map this onto the Active Inference framework: what multisensory prediction errors drove the body ownership update?
- Predict what would happen if the stroking were asynchronous (200ms delay between visual and tactile strokes). Why?

Write your response here

Part 2: Depersonalization Following Trauma

Goal: Analyze a clinical case of self-model attenuation.

Case: Patient S.L., 34, presents with persistent depersonalization following a car accident. She reports: "I feel like I'm watching myself from behind a glass wall. My hands don't feel like mine. Even my voice sounds like it belongs to a stranger."

- Interpret S.L.'s symptoms in Active Inference terms. Which component of the generative model is affected, and how?
- What role might precision play? (Hint: low precision on interoceptive prediction errors → attenuated self-model.)
- What therapeutic intervention might Active Inference suggest? (Consider interventions that increase interoceptive precision — e.g., body scanning, grounding exercises.)

Write your response here

Part 3: Body Integrity Identity Disorder

Goal: Analyze a rare condition through the lens of body representation.

Case: Patient R.M., 48, has felt since childhood that his left leg below the knee "does not belong" to him. He covers it with blankets, avoids looking at it, and has researched amputation. His leg is neurologically healthy.

- How does Active Inference explain this persistent mismatch between body and body model?
- Compare this case with phantom limb syndrome (missing limb, present model) — how are they inverse conditions?
- What does this case reveal about the relationship between the body and the body model? Which is more "real" for the patient?

Write your response here

Part 4: Interoception and Emotional Awareness

Goal: Connect interoceptive inference to emotional experience.

Case: Participant A.G. scores in the bottom 10th percentile on the Heartbeat Detection Task (counting heartbeats without taking a pulse). She also reports difficulty identifying her emotions (alexithymia). However, she shows normal autonomic responses (heart rate, skin conductance) to emotional stimuli.

- How does Active Inference explain the dissociation between normal autonomic responses and poor interoceptive awareness?
- If emotions are "predictions about interoceptive causes," what happens when interoceptive precision is low?
- Design a brief study to test whether interoceptive training improves emotional awareness.

Write your response here

Part 5: Synthesis

Write a 200-word synthesis explaining how all four cases illustrate the central claim: the self is not a given but an ongoing inference, vulnerable to disruption at multiple levels.

Write your response here

Lab Summary

Part	Skill Developed	Case Type
1	Experimental analysis	Body ownership illusion
2	Clinical reasoning	Depersonalization disorder
3	Comparative analysis	Body integrity identity disorder
4	Research design	Interoception and alexithymia
5	Integrative synthesis	Connecting four cases to the self-model framework